

Tanvir Rahman Anik

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RESEARCH INTERESTS

Medical Image Analysis · Explainable AI · Multimodal Learning · Generative AI · Natural Language Processing

RESEARCH EXPERIENCE

Research Collaborator

May 2024 – Present

College of Computer and Information Sciences, Jouf University, Saudi Arabia

Collaborator: Dr. Mohammad Azad | Remote Collaboration

- Researching an interpretable, parameter-efficient multimodal model for early Alzheimer's detection that fuses structural MRI, multi-omics molecular profiles, and cognitive records, handling missing modalities while keeping the model compact for resource-limited clinical settings.
- Conducted a systematic review of 68 studies on diffusion models in medical imaging and health informatics; analyzed model architectures, clinical applications, and explainability, identifying gaps in XAI adoption (22%) and clinician collaboration (57%) and proposing 10 evidence-based future research directions.
- Developed a stacked-ensemble model for knee osteoarthritis diagnosis, combining transfer-learning models (Xception, EfficientNetB5, InceptionV3) with explainable AI (Grad-CAM, Faster Score-CAM, LIME), achieving 96.93% accuracy in multi-class classification and 86.29% accuracy in binary classification.
- Co-authored a review of deep learning for knee osteoarthritis imaging, surveying pre-processing techniques, model architectures, and explainable AI frameworks.

PUBLICATIONS

Journal Articles

3. **Rahman Anik, T.**, & Azad, M. "Deep Learning in Knee Osteoarthritis: A Comprehensive Review of Imaging Modalities, Model Architectures, Grading Systems, and Explainable AI." (Submitted for review)
2. Azad, M., & **Rahman Anik, T.** "A Systematic Review of Diffusion Models for Medical Image-Based Diagnosis: Methods, Taxonomies, Clinical Integration, Explainability, and Future Directions" **Diagnostics** 🔗.
1. Azad, M., & **Rahman Anik, T.** "Automated Diagnosis of Knee Osteoarthritis: A Stacked Ensemble Deep Learning Approach with Explainable AI Techniques." **IEEE Access** 🔗.

Conference Publications

2. **Rahman Anik, T.**, Talukder, P., Faruki, I., Ibn Rahman, I. S., & Hossain, E. (2023). "Analysis of Automated Skin Disease Classification Exploiting Different Machine Learning Techniques." **IEEE 13th Annual Computing and Communication Workshop and Conference (CCWC)** 🔗, Las Vegas, USA.
1. Mouri, A. G., Talukder, P., **Rahman Anik, T.**, Ibn Rahman, I. S., Joy, S. K. S., & Shawon, M. T. R. (2022). "An Empirical Study on Bengali News Headline Categorization Leveraging Different Machine Learning Techniques." **25th International Conference on Computer and Information Technology (ICCIT)** 🔗, Cox's Bazar, Bangladesh.

EDUCATION

Ahsanullah University of Science and Technology

April 2018 – January 2023

Bachelor of Science in Computer Science and Engineering

CGPA: 3.378/4.00

Thesis: "Skin Disease Classification Using Different Machine Learning Techniques"

Supervisor: Md. Emam Hossain | Developed an automated skin disease classification system, achieving 99.33% accuracy with Xception on a dataset of 9,665 medical images.

Relevant Courses: Computer Architecture, Algorithms, Artificial Intelligence, Pattern Recognition, Digital Image Processing, Computer Networks, and Operating Systems.

INDUSTRY EXPERIENCE

Software Engineer

April 2026 to Present

Shottify (shottify.org) 🔗

Live AI-powered news verification and fact-checking platform

- Redesigned the platform's X (Twitter) data-collection and search pipeline and engineered the LLM prompts for relevance filtering, cutting a full scrape run from 2–3 hours to 15–20 minutes (~85% faster) and reducing irrelevant content from ~95% to under 5%.
- Led a full OWASP ASVS L2 security assessment (automated SAST/SCA/secret/container scans plus manual auth, IDOR, SSRF, and LLM prompt-injection testing); remediated critical bugs, and hardened the Cloudflare edge (WAF, rate limiting, origin-lock).
- Mentored 3 junior engineers through task planning, work assignment, and code reviews, and collaborated closely with the content team, stakeholders, and project managers to shape product decisions and delivery.

LLM Engineer (Contractual)

June 2026

Xynoptik, Australia | Remote

Teer — production LLM/RAG platform for a B2B advertising & media marketplace

- Profiled latency of a production LangGraph RAG agent (FastAPI on AWS ECS), instrumenting per-node timing to CloudWatch and isolating the two nodes responsible for ~95% of query latency to direct optimization, and fixed it by loading only the tables each query needs, sharply reducing query time.
- Built an automated evaluation suite of 100 role-based test questions scored against expected answers, turning prompt changes into a measurable pass rate(95%), and added per-question leakage guardrails (deny-list assertions) that automatically flag any cross-tenant data exposure.
- Migrated the platform’s AI prompts to a versioned management system (Langfuse) so they could be updated without a code release, and parallelized pipeline steps to speed up responses.

Machine Learning Engineer

May 2025 to January 2026

Interactive Cares, Dhaka, Bangladesh

- Engineered an enterprise AI interview platform on a scalable microservices architecture (OpenAI, Gemini, Redis caching), reducing candidate screening time by 70%.
- Built a CV screening system (FastAPI, AWS) that scores and ranks 500+ candidates in minutes, cutting manual review effort by 80%.
- Developed a responsive Next.js web app with role-based access (candidates, HR, guests) and real-time video interview interfaces.

Machine Learning Engineer

July 2023 to April 2025

Bista Solutions Inc., Dhaka, Bangladesh

- Designed RAG-based AI assistants (Django, LangChain, ChromaDB) enabling natural-language ERP operations across inventory, orders, and finance.
- Fine-tuned a Meta LLaMA 3.1 (8B) model for text-to-SQL over ERP data (85.48% accuracy), making complex data accessible to non-technical users.
- Built LSTM sales-forecasting models ($R^2 = 0.87$) integrated into Odoo dashboards to improve inventory planning.

TECHNICAL SKILLS

Programming Languages	Python, C, C++ , Java, SQL, HTML, CSS, JavaScript.
Generative AI & LLM Tools	LangChain, LangSmith (observability, tracing, evaluation), HuggingFace Transformers, OpenAI Assistants API, Retrieval-Augmented Generation (RAG), Vector databases.
Frameworks & Libraries	TensorFlow, PyTorch, Keras, Scikit-learn, OpenCV, FastAPI, Django, Flask, Pandas, NumPy, Matplotlib, Seaborn, Android, Selenium WebDriver.
Machine Learning	Supervised & unsupervised learning, Regression, Classification, Clustering, Ensemble Learning, Model Evaluation & Interpretability, CNN, DNN, Transfer Learning, Transformers.
Databases	MySQL, PostgreSQL, Oracle.
Cloud & DevOps Tools	Git/GitHub, Docker, AWS, CI/CD Pipelines, Nginx, Virtualization (VMware), Linux, Cloud-based deployment.
Others	LaTeX, Jupyter notebook, Arduino, Proteus, Microsoft Office, Adobe Photoshop, Adobe After Effects.

STANDARDIZED TESTS

IELTS (December 2024): **Overall 7.0** (Listening: 8.0, Reading: 6.0, Writing: 7.0, Speaking: 6.5)

SELECTED PROJECTS

- 2D-to-3D Model Generator | Python, PyTorch, Transformers, Computer Vision
2D images to 3D models using Stability AI’s TripoSR with automated mesh generation and .glb model export .
- NewsDocument-Classifer | PyTorch, Bi-LSTM, GloVe, Multi-Head Attention, Scikit-learn
Hierarchical Attention Network for multi-class text classification with dual-stage attention, residual connections, and gating mechanism.
- AI-Interviewer | Next.js, TypeScript, FastAPI, OpenAI+Gemini, Redis, MySQL, Docker
Conversational interview platform with microservices architecture and multimodal processing via OpenAI+Gemini APIs.
- CV-Sorter | FastAPI, OpenAI APIs, MySQL, AWS S3
AI-powered CV screening platform with async batch processing and intelligent candidate ranking.

LLM Fine-tuning for Text-to-SQL | *Meta-LLaMA 3.1, QLoRA, HuggingFace*

Fine-tuned 8B parameter model using QLoRA with 4-bit quantization, achieving 85.48% exact match accuracy.

Acumen Helpbot | *Django, ChromaDB, Sentence Transformers*

RAG-based chatbot with vector search and embeddings-based retrieval for document Q&A.

Odoo-Copilot | *Django, OpenAI Assistants API, LangChain, ChromaDB*

Agentic ERP chatbot enabling natural-language queries and workflow automation through multi-LLM orchestration.

Knee Osteoarthritis Severity Prediction | *TensorFlow, Keras, OpenCV*

Stacked ensemble model for KOA classification from X-rays with explainability via Grad-CAM, Score-CAM, and LIME.

TEACHING EXPERIENCE

Instructor

2018 – 2022

Aid Academy, Science Coaching Center, Dhaka

- Delivered structured lectures in biology and chemistry to high school and college students.
- Provided personal academic assistance and guidance to students preparing for board exams.
- Designed curricula, created exam materials, conducted assessments, and evaluated student performance.

HONORS AND PROFESSIONAL ACTIVITIES

- Second runner-up, Software Project Showcase, AUST CSE Week 2022.
- Participated in AUST Programming Contests (2018-2021), Soccer Bot & Line Follower Robot competitions.
- Organizing committee member for departmental events, member of AUST Innovation & Design Club (2021-2022),